Module 07 Lab - Better Model Evaluation**Objective:** To move beyond simple accuracy and learn how to use more sophisticated and reliable evaluation techniques, including the confusion matrix, precision, recall, and cross-validation.**In this lab, you will write the code to generate and interpret these advanced evaluation metrics.**

Part 1: Setup and Model TrainingLet's first train a model so we have something to evaluate. We will use the Titanic dataset again to predict survival.

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

# Load the Titanic dataset
df = pd.read_csv(
    'https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv'
)

# Convert Sex into numbers
df['Sex'] = df['Sex'].map({'male': 0, 'female': 1})

# Fill missing ages with the median age
df['Age'] = df['Age'].fillna(df['Age'].median())

# Select features and target
features = ['Age', 'Pclass', 'Sex', 'Fare']
target = 'Survived'

X = df[features]
y = df[target]

# Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train the logistic regression model
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

# Predict survival on the test set
y_pred = model.predict(X_test)

print("Model training completed successfully!")
print("Training samples:", len(X_train))
print("Testing samples:", len(X_test))
```

```
Model training completed successfully!
Training samples: 712
Testing samples: 179
```

Part 2: The Confusion Matrix**Concept:** A confusion matrix gives you a more detailed breakdown of a model's performance than accuracy alone. It's a table that shows you where your model got things right and where it got them wrong.It has four quadrants:* **True Positives (TP):** Correctly predicted positive (e.g., predicted survival, and they did survive).* **True Negatives (TN):** Correctly predicted negative (e.g., predicted did not survive, and they did not).* **False Positives (FP):** Incorrectly predicted positive (e.g., predicted survival, but they did not). Also

called a "Type I Error".* **False Negatives (FN):** Incorrectly predicted negative (e.g., predicted did not survive, but they did). Also called a "Type II Error".

Task 1: Generate and Plot a Confusion Matrix**Your Task:** Use `confusion_matrix` from `sklearn.metrics` to calculate the matrix and `seaborn.heatmap` to visualize it.

```
from sklearn.metrics import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

# Calculate the confusion matrix
cm = confusion_matrix(y_test, y_pred)

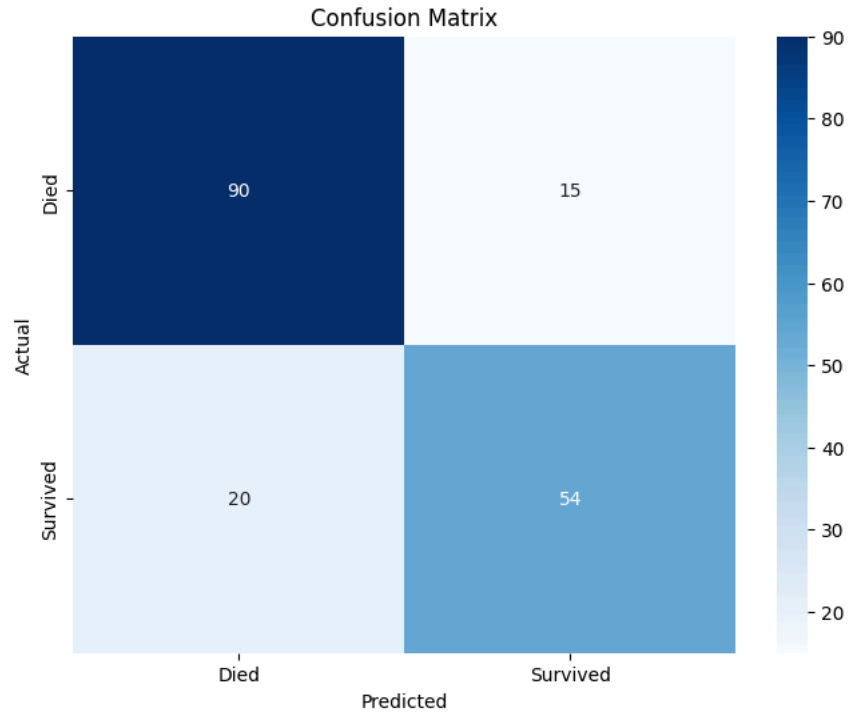
# Display the confusion matrix
print("Confusion Matrix:")
print(cm)

# Plot the confusion matrix
plt.figure(figsize=(8, 6))

sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues',
    xticklabels=['Died', 'Survived'],
    yticklabels=['Died', 'Survived']
)

plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
```

```
Confusion Matrix:
[[90 15]
 [20 54]]
```



Part 3: Precision, Recall, and F1-Score**Concept:** From the confusion matrix, we can calculate more nuanced metrics:* **Precision:** Of all the times the model predicted **positive**, how often was it correct? * Formula: `TP / (TP + FP)` * *Use Case:* When the cost of a **False Positive** is high. (e.g., a spam filter; you don't want to incorrectly mark an important email as spam).* **Recall (Sensitivity):** Of all the actual **positives**, how many did the model correctly identify? * Formula: `TP / (TP + FN)` * *Use Case:* When the cost of a **False Negative** is high. (e.g., a medical test for a serious disease; you don't want to miss a real case).* **F1-Score:** The harmonic mean of Precision and Recall. It provides a single score that balances both.

Task 2: Generate a Classification Report**Your Task:** Use `classification_report` from `sklearn.metrics` to get a summary of these metrics for each class.

```
from sklearn.metrics import classification_report

# Generate the classification report
report = classification_report(
    y_test,
    y_pred,
    target_names=['Died', 'Survived']
)

# Print the results
print("Classification Report:")
print(report)
```

```
Classification Report:
              precision    recall  f1-score   support

        Died       0.82      0.86      0.84       105
    Survived       0.78      0.73      0.76        74

    accuracy                           0.80       179
   macro avg       0.80      0.79      0.80       179
weighted avg       0.80      0.80      0.80       179
```

Part 4: Cross-Validation**Concept:** A single train-test split can be lucky or unlucky. What if, by chance, all the "easy" examples ended up in our test set? Our accuracy score would be misleadingly high.**Cross-Validation (CV)** solves this. It splits the data into multiple "folds" (e.g., 5 or 10). It then trains and evaluates the model multiple times, using a different fold as the test set each time. The final score is the average of the scores from all folds.This gives a much more robust and reliable estimate of the model's true performance.

Task 3: Perform Cross-Validation**Your Task:** Use `cross_val_score` from `sklearn.model_selection` to perform 5-fold cross-validation on your model.

```
from sklearn.model_selection import cross_val_score

# Perform 5-fold cross-validation
cv_scores = cross_val_score(
    model,
    X,
    y,
    cv=5
)
```

```
# Display the results
print(f"Scores for each fold: {cv_scores}")
print(f"Average CV Score: {cv_scores.mean():.2%}")
print(f"Standard Deviation of CV Scores: {cv_scores.std():.4f}")
```

```
Scores for each fold: [0.7877095  0.78089888 0.78651685 0.7752809  0.80337079]
Average CV Score: 78.68%
Standard Deviation of CV Scores: 0.0094
```

Knowledge Check**Instructions:** Answer the following questions in this markdown cell.

1. **Describe a real-world scenario where you would care more about a model's Precision than its Recall.** A good example is an email spam filter. I would want high precision because important emails should not be marked as spam by mistake. Missing a few spam emails might be annoying, but losing an important email could cause a bigger problem.
2. **Describe a real-world scenario where you would care more about a model's Recall than its Precision.** A good example is a medical screening system that checks for serious diseases. I would want high recall because missing someone who actually has the disease could be dangerous. It is better to identify more possible cases and have doctors check them than to miss someone who needs help.
3. **Why is a cross-validation score generally more trustworthy than a score from a single train-test split?****[ENTER YOUR ANSWERS HERE]** Cross-validation is more trustworthy because it tests the model using different parts of the dataset. A single train-test split might give a high or low score just by chance. By using five folds and averaging the results, we get a better idea of how consistently the model performs.